C6.2 How do chemists control the rate of reactions?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*
Controlling rate of reaction enables industrial chemists to optimise the rate at which a chemical product can be made safely. The rate of a reaction can be altered by altering conditions such as temperature, concentration, pressure and surface area. A model of particles colliding helps to explain why and how each of these factors affects rate; for example, increasing the temperature increases the rate of collisions and, more significantly, increases the energy available to the particles to overcome the activation energy and react. A catalyst increases the rate of a reaction but can be recovered, unchanged, at the end. Catalysts work by providing an alternative route for a reaction with a lower activation energy. Energy changes for uncatalysed and catalysed reactions have different reaction profiles. The use of a catalyst can reduce the economic and environmental cost of an industrial process, leading to more sustainable 'green' chemical processes.	1. describe the effect on rate of reaction of changes in temperature, concentration, pressure, and surface area 2. explain the effects on rates of reaction of changes in temperature, concentration and pressure in terms of frequency and energy of collision between particles 3. explain the effects on rates of reaction of changes in the size of the pieces of a reacting solid in terms of surface area to volume ratio 4. describe the characteristics of catalysts and their effect on rates of reaction 5. identify catalysts in reactions 6. explain catalytic action in terms of activation energy

Linked learning opportunities

Specification links

- Endothermic and exothermic reactions and energy level diagrams. (C1)

Practical Work:

- Investigate the effect of temperature and concentration on rate of reactions.
- Compare methods of following rate

Ideas about Science:

- Use the particle model to explain factors that affect rates of reaction (IaS3)
- The use of catalysts supports more sustainable industrial processes.(IaS4)

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Rate of reaction can be determined by measuring the rate at which a product is made or the rate at which a reactant is used. **Some reactions involve a colour change or form a solid in a solution; the rate of these reactions can be measured by timing the changes that happen in the solutions by eye or by using apparatus such as a colorimeter**. Reactions that make gases can be followed by measuring the volume of gas or the change in mass over time. On graphs showing the change in a variable such as concentration over time, the gradient of a tangent to the curve is an indicator of rate of change at that time. The average rate of a reaction can be calculated from the time taken to make a fixed amount of product.	7. suggest practical methods for determining the rate of a given reaction including*: PAG8* for reactions that produce gases: i. gas syringes or collection over water can be used to measure the volume of gas produced ii. mass change can be followed using a balance **measurement of physical factors:** **iii. colour change** **iv. formation of a precipitate** 8. interpret rate of reaction graphs M4a, M4b **9. interpret graphs of reaction conditions versus rate *(separate science only)*** M4a, M4b ⓘ *an understanding of orders of reaction is not required* 10. use arithmetic computation and ratios when measuring rates of reaction M1a, M1c 11. draw and interpret appropriate graphs from data to determine rate of reaction M2b, M4b, M4c 12. determine gradients of graphs as a measure of rate of change to determine rate M4b, M4d, M4e	**Practical work:** • Designing and carrying out investigations into rates. Analysing and interpreting data. Use of apparatus to make measurements. Use of heating equipment. Safe handling of chemicals. Measuring rates of reaction using two different methods **Specification link:** • enzymes in biological processes (B3.1)

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Enzymes are proteins that catalyse processes in living organisms. They work at their optimum within a narrow range of temperature and pH. Enzymes can be adapted and sometimes synthesised for use in industrial processes. Enzymes limit the conditions that can be used but this can be an advantage because if a process can be designed to use an enzyme at a lower temperature than a traditional process, this reduces energy demand.	13.	use proportionality when comparing factors affecting rate of reaction M1c	
	14.	describe the use of enzymes as catalysts in biological systems and some industrial processes	